Switch-Case

Choosing between many fixed values with switch, case, break and default

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What is switch-case?

`switch` compares one expression against a list of constant values. When a `case` matches, the code under it runs. It is a cleaner alternative to a long `if / else if` chain when you are checking a single variable for equality.

Syntax

```
switch (expression) {
    case value1:
        // runs when expression == value1
        break;
    case value2:
        // runs when expression == value2
        break;
    default:
        // runs when no case matches
}
```

Key rules

- **switch (expression)**: the expression must be an integer type. `char` also works, because characters are stored as their ASCII codes.
- **case value:** only checks *equality* with a constant. You cannot write `case x > 5:`; ranges and comparisons need `if`.
- **break;** ends the switch. Without it, execution continues into the next case. This is called **fall-through**.
- **default:** runs when no case matches, like the final `else` of an if chain. It is optional and can be placed anywhere, although it is usually written last.
- Character cases are case-sensitive: `'a'` and `'A'` are different values.

EXAMPLE 1

Month number to month name

Print the name of the month for a given month number (1-12).

CODE

```c
#include <stdio.h>

int main() {
    int month = 5;

    switch (month) {
        case 1:  printf("January\n");   break;
        case 2:  printf("February\n");  break;
        case 3:  printf("March\n");     break;
        case 4:  printf("April\n");     break;
        case 5:  printf("May\n");       break;
        case 6:  printf("June\n");      break;
        case 7:  printf("July\n");      break;
        case 8:  printf("August\n");    break;
        case 9:  printf("September\n"); break;
        case 10: printf("October\n");   break;
        case 11: printf("November\n");  break;
        case 12: printf("December\n");  break;
        default: printf("Invalid month\n");
    }
    return 0;
}
```

SAMPLE RUN

```
May
```

EXAMPLE 2

Seasons with fall-through

Read a month number and print its season. Several cases share the same code by leaving out `break` between them (intentional fall-through).

CODE

```c
#include <stdio.h>

int main() {
    int month;

    printf("Which month is it (1-12)? ");
    scanf("%d", &month);

    switch (month) {
        case 12:
        case 1:
        case 2:
            printf("Winter\n");
            break;
        case 3:
        case 4:
        case 5:
            printf("Spring\n");
            break;
        case 6:
        case 7:
        case 8:
            printf("Summer\n");
            break;
        case 9:
        case 10:
        case 11:
            printf("Autumn\n");
            break;
        default:
            printf("Invalid month number!\n");
    }
    return 0;
}
```

SAMPLE RUN

```
Which month is it (1-12)? 10
Autumn
```

EXAMPLE 3

Simple calculator

Read two integers and an operator (+, -, *, /). Perform the chosen operation and print the result. Division by zero is handled separately.

CODE

```c
#include <stdio.h>

int main() {
    int num1, num2;
    char op;

    printf("Enter the first number: ");
    scanf("%d", &num1);
    printf("Enter the second number: ");
    scanf("%d", &num2);
    printf("Enter an operation (+, -, *, /): ");
    scanf(" %c", &op);      /* the space skips the leftover newline */

    switch (op) {
        case '+':
            printf("Result: %d\n", num1 + num2);
            break;
        case '-':
            printf("Result: %d\n", num1 - num2);
            break;
        case '*':
            printf("Result: %d\n", num1 * num2);
            break;
        case '/':
            if (num2 != 0) {
                printf("Result: %d\n", num1 / num2);
            } else {
                printf("Error: division by zero is not allowed!\n");
            }
            break;
        default:
            printf("Invalid operation!\n");
    }
    return 0;
}
```

SAMPLE RUN

```
Enter the first number: 20
Enter the second number: 4
Enter an operation (+, -, *, /): /
Result: 5
```

EXAMPLE 4

Vowel or consonant

Read a letter and print whether it is a vowel or a consonant. Upper- and lower-case vowels are grouped with fall-through.

CODE

```c
#include <stdio.h>

int main() {
    char ch;

    printf("Enter a letter: ");
    scanf(" %c", &ch);

    switch (ch) {
        case 'a': case 'e': case 'i': case 'o': case 'u':
        case 'A': case 'E': case 'I': case 'O': case 'U':
            printf("Vowel\n");
            break;
        default:
            printf("Consonant\n");
            break;
    }
    return 0;
}
```

SAMPLE RUN

```
Enter a letter: E
Vowel
```

EXAMPLE 5

Greeting by hour (switch inside a function)

Read an hour between 0 and 23 and print a greeting: 0-5 Good night, 6-17 Good day, 18-23 Good evening. A function named `greet` must use switch-case.

CODE

```c
#include <stdio.h>

void greet(int hour) {
    switch (hour / 6) {         /* 0-5 -> 0, 6-11 -> 1, 12-17 -> 2, 18-23 -> 3 */
        case 0:
            printf("Good night!\n");
            break;
        case 1:
        case 2:
            printf("Good day!\n");
            break;
        case 3:
            printf("Good evening!\n");
            break;
        default:
            printf("Invalid hour\n");
            break;
    }
}

int main() {
    int hour;

    printf("What time is it (0-23)? ");
    scanf("%d", &hour);

    if (hour < 0 || hour > 23) {
        printf("Invalid hour\n");
    } else {
        greet(hour);
    }
    return 0;
}
```

SAMPLE RUN

```
What time is it (0-23)? 19
Good evening!
```

Note. Since switch cannot test ranges, the trick `hour / 6` turns each 6-hour block into a single integer that a case can match. The original exercise text used the Turkish greeting "Good morning" for 0-5; it is written here as "Good night", which fits those hours better in English.

EXAMPLE 6

Letter grade from a score

Read a score (0-100) and print the letter grade with a function `letterGrade` using switch-case: 90-100 A, 80-89 B, 70-79 C, 60-69 D, below 60 F.

CODE

```c
#include <stdio.h>

void letterGrade(int score) {
    switch (score / 10) {
        case 10:
        case 9:
            printf("Letter grade: A\n");
            break;
        case 8:
            printf("Letter grade: B\n");
            break;
        case 7:
            printf("Letter grade: C\n");
            break;
        case 6:
            printf("Letter grade: D\n");
            break;
        default:
            printf("Letter grade: F\n");
            break;
    }
}

int main() {
    int score;

    printf("Enter the student's score (0-100): ");
    scanf("%d", &score);

    if (score < 0 || score > 100) {
        printf("Invalid score\n");
    } else {
        letterGrade(score);
    }
    return 0;
}
```

SAMPLE RUN

```
Enter the student's score (0-100): 84
Letter grade: B
```

EXAMPLE 7

Bit shifting menu

Read a number and a menu choice. Choice 1 shifts the number one bit to the left (multiply by 2); choice 2 shifts it one bit to the right (integer divide by 2).

CODE

```c
#include <stdio.h>

int main() {
    int num, choice;

    printf("Enter a number: ");
    scanf("%d", &num);
    printf("1: Shift left\n2: Shift right\nChoose an operation: ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            printf("Result (shift left): %d\n", num << 1);
            break;
        case 2:
            printf("Result (shift right): %d\n", num >> 1);
            break;
        default:
            printf("Invalid choice!\n");
            break;
    }
    return 0;
}
```

SAMPLE RUN

```
Enter a number: 17
1: Shift left
2: Shift right
Choose an operation: 2
Result (shift right): 8
```

EXAMPLE 8

Positive, negative or zero (nested switch)

Classify a number using switch only. A comparison like `num > 0` evaluates to 1 (true) or 0 (false), so it can be used as the switch expression.

CODE

```c
#include <stdio.h>

void checkNumber(int num) {
    switch (num > 0) {
        case 1:
            printf("The number %d is positive.\n", num);
            break;
        case 0:
            switch (num < 0) {
                case 1:
                    printf("The number %d is negative.\n", num);
                    break;
                case 0:
                    printf("The number is zero.\n");
                    break;
            }
            break;
    }
}

int main() {
    checkNumber(5);
    checkNumber(-3);
    checkNumber(0);
    return 0;
}
```

SAMPLE RUN

```
The number 5 is positive.
The number -3 is negative.
The number is zero.
```

EXAMPLE 9

Character type with case ranges

Decide whether a character is an uppercase letter, a lowercase letter, a digit, or something else.

CODE

```c
#include <stdio.h>

void charType(char c) {
    switch (c) {
        case 'A' ... 'Z':
            printf("'%c' is an uppercase letter.\n", c);
            break;
        case 'a' ... 'z':
            printf("'%c' is a lowercase letter.\n", c);
            break;
        case '0' ... '9':
            printf("'%c' is a digit.\n", c);
            break;
        default:
            printf("'%c' is a special character or whitespace.\n", c);
    }
}

int main() {
    charType('B');
    charType('z');
    charType('5');
    charType('@');
    return 0;
}
```

SAMPLE RUN

```
'B' is an uppercase letter.
'z' is a lowercase letter.
'5' is a digit.
'@' is a special character or whitespace.
```

Note. The `case 'A' ... 'Z':` range syntax is a GCC/Clang extension, not standard C. It will not compile with MSVC; in portable code use `if (c >= 'A' && c <= 'Z')` instead.